

24/04/2026.

## SQL Aggregate Functions and Operators.

① LIST ALL DISTINCT DEPARTMENTS IN THE STUDENTS TABLE

SELECT DISTINCT department  
FROM students\_table;

| Departments |
|-------------|
| IT          |
| HR          |
| FINANCE     |

② GET THE AVERAGE AGE OF STUDENTS PER DEPARTMENT

SELECT department, AVG(age) AS average  
FROM students  
GROUP BY department;

| Department | Average - age |
|------------|---------------|
| IT         | 20.5          |
| HR         | 22.0          |
| FINANCE    | 23.0          |

③ SHOW DEPARTMENTS WITH MORE THAN 1 STUDENT.

SELECT department, COUNT(student\_id)  
FROM students  
GROUP BY department

HAVING COUNT(student\_id) > 1;

| department | student_count |
|------------|---------------|
| IT         | 2             |
| HR         | 2             |

- ④ GET ALL STUDENTS WHERE AGE IS BETWEEN 21 and 23  
 SELECT \*  
 FROM students\_Table  
 WHERE age BETWEEN 21 and 23;

| student_id | name    | age | department |
|------------|---------|-----|------------|
| 2          | Bob     | 22  | HR         |
| 3          | Charlie | 21  | IT         |
| 4          | Diana   | 23  | FINANCE    |
| 5          | Eve     | 22  | HR         |

- ⑤ LIST ALL STUDENTS IN THE IT OR HR DEPARTMENT WHO ARE OLDER THAN 21.

SELECT \*  
 FROM students  
 WHERE department IN ('IT', 'HR')  
 AND age > 21;

| student_id | name | age | department |
|------------|------|-----|------------|
| 2          | Bob  | 22  | HR         |
| 5          | Eve  | 22  | HR         |

### ⑥ THE COURSES TABLE

- ⑥ SHOW TOTAL CREDITS PER DEPARTMENT, ONLY FOR DEPARTMENTS WITH MORE THAN 5 TOTAL CREDITS.

SELECT department, SUM(credits) AS total\_credits  
 FROM courses  
 GROUP BY department  
 HAVING SUM(credits) > 5



| department | total_credits |
|------------|---------------|
| IT         | 11            |

⑦ LIST ALL COURSES THAT DO NOT HAVE 4 CREDITS.

SELECT \*

FROM courses

WHERE credits  $\neq$  4;

| Course_id | Course_name | department | Credits |
|-----------|-------------|------------|---------|
| 101       | SQL Basics  | IT         | 3       |
| 104       | Excel       | Finance    | 2       |
| 105       | Statistics  | HE         | 3       |

⑧ SHOW THE TOP 3 courses by credits in descending order.

SELECT course\_name, department, credit

FROM courses

ORDER BY credits DESC

LIMIT 3;

| Course_name  | department | credits |
|--------------|------------|---------|
| Python       | IT         | 4       |
| Data Science | IT         | 4       |
| SQL Basics   | IT         | 3       |

THE ENROLLMENTS TABLE

⑨ GET THE MAXIMUM, MINIMUM AND AVERAGE GRADE ACROSS ALL ENROLLMENTS.

SELECT

MAX(grade) AS max\_grade,

MIN(grade) AS min\_grade,

AVG(grade) AS average\_grade

FROM enrollments\_Table

| Max_grade | min_grade | avg_grade |
|-----------|-----------|-----------|
| 90        | 78        | 84.6      |

⑩ HOW MANY ENROLLMENTS EXIST PER COURSE

```
SELECT course_id, COUNT(enrollment_id) AS enrollment_count
FROM enrollments
GROUP BY course_id
```

| Course_id | enrollment_count |
|-----------|------------------|
| 101       | 1                |
| 102       | 1                |
| 103       | 1                |
| 104       | 1                |
| 105       | 1                |

THE SALARIES TABLE

⑪ FIND TOTAL SALARY AND TOTAL BONUS PER DEPARTMENT.

```
SELECT
  department
  SUM(salary) AS total_salary,
  SUM(bonus) AS total_bonus
FROM employees
GROUP BY department;
```

| department | total_salary | total_bonus |
|------------|--------------|-------------|
| IT         | 122,000      | 10,500      |
| HR         | 109,000      | 7,500       |
| Finance    | 70,000       | 6,000       |



- ② SHOW DEPARTMENTS WHERE AVERAGE SALARY IS ABOVE 55 000
- ```
SELECT department, AVG(salary) AS avg
FROM employees
GROUP BY department
HAVING AVG(salary) > 55 000;
```

| department | avg_salary |
|------------|------------|
| Finance    | 70 000     |
| IT         | 61 000     |

- ③ LIST EMPLOYEES WHOSE SALARY PLUS BONUS IS GREATER THAN 60 000

```
SELECT
employee_id, name, salary, bonus (salary + bonus) AS total_
compens FROM employees WHERE (salary + bonus) > 60 000;
```

| employee_id | name  | salary | bonus |
|-------------|-------|--------|-------|
| 1           | Tom   | 60 000 | 5000  |
| 3           | Spike | 70 000 | 6000  |
| 4           | Tyke  | 62 000 | 5500  |

- ④ THE PROJECTS TABLE.

SHOW TOTAL AND AVERAGE BUDGET PER DEPARTMENT. ONLY  
INCLUDE DEPARTMENTS WITH AVERAGE BUDGET ABOVE 70 000

```
SELECT
```

- ⑤
- ```
depart, SUM(Budget) AS total_budget,
AVG(budget) AS average_budget
FROM projects_table
GROUP BY department,
HAVING AVG(budget) > 70 000;
```

| department | total budget | average - budget |
|------------|--------------|------------------|
| Finance    | 80 000       | 80 000           |
| IT         | 270 000      | 135 000          |

Q5) LIST ALL PROJECTS WITH BUDGETS BETWEEN 50 000 AND 120 000, excluding THE MARKETING DEPARTMENT.

SELECT \*

FROM projects\_table

WHERE budget BETWEEN 50 000 AND 120 000  
AND department <> 'Marketing' ;

| Project Id | Project_name   | department | Budget  |
|------------|----------------|------------|---------|
| 1          | AI App         | IT         | 120 000 |
| 2          | PAYROLL SYSTEM | Finance    | 80 000  |
| 5          | HIZ Portal     | IT         | 50 000  |